



Lab Assignment – 11 (Basic Indexing Concepts)

Department: CSE

Subject: DIS Lab

Code: 22CPP217

Objective: Understand the importance of indexing in databases, implement basic indexing techniques using SQL, and analyze different indexing strategies such as ordered indices, dense vs sparse indexing, multilevel indexing, and secondary indices for efficient query processing.

Pre-Lab Setup (Indexing Environment)

SQL

-- 0. Reset Environment

SET SQL_SAFE_UPDATES = 0;

DROP TABLE IF EXISTS Student;

DROP TABLE IF EXISTS Course;

DROP TABLE IF EXISTS Enrollment;

-- 1. Create Schema

```
CREATE TABLE Student (  
    Student_ID INT PRIMARY KEY,  
    Name VARCHAR(50),  
    Dept_Name VARCHAR(30)  
);
```

```
CREATE TABLE Course (  
    Course_ID INT PRIMARY KEY,  
    Course_Name VARCHAR(50),  
    Credits INT  
);
```

```
CREATE TABLE Enrollment (  
    Enroll_ID INT PRIMARY KEY,  
    Student_ID INT,  
    Course_ID INT,  
    Semester VARCHAR(10),  
    Grade CHAR(1)  
);
```

```
-- 2. Insert Sample Data
INSERT INTO Student VALUES
(101, 'Aarav', 'CSE'),
(102, 'Neha', 'CSE'),
(103, 'Rohan', 'Physics'),
(104, 'Meera', 'Math'),
(105, 'Ishita', 'History');

INSERT INTO Course VALUES
(201, 'DBMS', 4),
(202, 'OS', 3),
(203, 'Physics', 4),
(204, 'Algebra', 3);

INSERT INTO Enrollment VALUES
(1, 101, 201, 'Sem1', 'A'),
(2, 101, 202, 'Sem1', 'B'),
(3, 102, 201, 'Sem1', 'A'),
(4, 103, 203, 'Sem1', 'C'),
(5, 104, 204, 'Sem1', 'A'),
(6, 102, 202, 'Sem1', 'B');
```

Q1: Basic Index Creation & Usage

- Retrieve the student record with Student_ID = 101.
- Create an index on Student_ID:
- Execute the same query again after index creation.
- Display all indexes of the Student table:

Q2: Index on Non-Key Attribute

- Create an index on Dept_Name:
- Retrieve all students from 'CSE' department.
- Count number of students per department:

Q3: Index on Course Table

- Create an index on Course_Name:
- Retrieve course details where Course_Name = 'DBMS'.
- Retrieve courses having Credits ≥ 3 .

Q4: Composite Index (Multiple Columns)

- Create a composite index on the Enrollment table:
- Retrieve records where:
- Retrieve all courses taken by Student_ID = 102.

Q5: Index Maintenance (Drop & Recreate)

- Drop index on Dept_Name:
- Verify index removal:
- Recreate the index again.

Q6: Multiple Index Usage

- Create indexes on:
 - Student(Name)
 - Enrollment(Grade)
- Retrieve:
 - Students whose Name starts with 'A'
 - Students with Grade = 'A'

Q7: Combined Query Optimization

- Write a query to display:
 - Student Name
 - Course Name
 - Grade
- Ensure indexes exist on:
 - Student_ID
 - Course_ID

SQL Reference (Hints)

SQL

```
CREATE INDEX index_name ON table_name(column_name);
```

```
CREATE INDEX index_name  
ON table_name(col1, col2);
```

```
DROP INDEX index_name ON table_name;
```

```
SHOW INDEX FROM table_name;
```